Homework #1: Prove the Gradients of the Sigmoid and Tanh Functions

In this exercise we derive the gradients of two common activation functions in deep learning: the **sigmoid** function and the **tanh** function.

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1. Sigmoid Function Gradient

The sigmoid function is defined as:

$$\sigma(x)=\frac{1}{1+e^{-x}}$$

Goal

Show that:

$$\frac{d\,\sigma(x)}{d\,x}=\sigma(x)(1-\sigma(x))$$

Proof

We rewrite $\sigma(x)$ as a power:

$$\sigma(x)=¿$$

Let $u(x)=1+e^{-x}$, so $u'(x)=-e^{-x}$.

Applying the chain rule:

$$\frac{d\,\sigma}{d\,x}=-u(x)^{-2}\cdot u'(x)=-\frac{1}{(1+e^{-x})^2}\cdot(-e^{-x})=\frac{e^{-x}}{(1+e^{-x})^2}$$

Now we express this in terms of $\sigma(x)$. Notice that:

$$\sigma(x)=\frac{1}{1+e^{-x}}\text{ and }1-\sigma(x)=\frac{e^{-x}}{1+e^{-x}}$$

Therefore:

$$\sigma(x)(1-\sigma(x))=\frac{1}{1+e^{-x}}\cdot\frac{e^{-x}}{1+e^{-x}}=\frac{e^{-x}}{(1+e^{-x})^2}$$

We conclude:

$$\frac{d\,\sigma(x)}{d\,x} = \sigma(x)(1-\sigma(x))$$

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2. Tanh Function Gradient

The hyperbolic tangent function is defined as:

$$\tanh(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$$

Goal

Show that:

$$\frac{d\tanh(x)}{d\,x} = 1-\tanh^{2}(x)$$

Proof

Let:

$$a(x) = e^{x}-e^{-x}, b(x) = e^{x}+e^{-x}$$

so that $\tanh(x) = \frac{a(x)}{b(x)}$, with:

$$a'(x) = e^{x}+e^{-x} = b(x), b'(x) = e^{x}-e^{-x} = a(x)$$

Applying the **quotient rule**:

$$\frac{d\tanh(x)}{d\,x} = \frac{a'(x)b(x)-a(x)b'(x)}{b(x)^{2}} = \frac{b(x)^{2}-a(x)^{2}}{b(x)^{2}}$$

Expanding the numerator:

$$b(x)^{2} = (e^{x}+e^{-x})^{2} = e^{2x}+2+e^{-2x}$$

$$a(x)^{2} = (e^{x}-e^{-x})^{2} = e^{2x}-2+e^{-2x}$$

$$b(x)^{2}-a(x)^{2} = 4$$

Therefore:

$$\frac{d\tanh(x)}{d\,x} = \frac{4}{(e^{x}+e^{-x})^{2}}$$

Now observe that:

$$1-\tanh^2(x)=1-\frac{\left(e^x-e^{-x}\right)^2}{\left(e^x+e^{-x}\right)^2}=\frac{\left(e^x+e^{-x}\right)^2-\left(e^x-e^{-x}\right)^2}{\left(e^x+e^{-x}\right)^2}=\frac{4}{\left(e^x+e^{-x}\right)^2}$$

We conclude:

$$\frac{d\tanh(x)}{dx}=1-\tanh^2(x)$$

Summary

Function	Definition	Gradient
$\sigma(x)$	$\frac{1}{1+e^{-x}}$	$\sigma(x)(1-\sigma(x))$
$\tanh(x)$	$\frac{e^x-e^{-x}}{e^x+e^{-x}}$	$1-\tanh^2(x)$

Both gradients can be expressed in terms of the function itself — this is what makes them computationally efficient in backpropagation.